

CS1 Week 8

Annotated

Time-Domain Specifications, PID Control



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Recap Last Week

Quiz 1

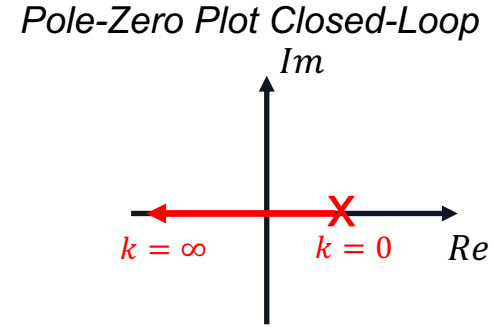
The Root Locus plot is a plot that shows...

- A.** that shows us how a closed-loop system behaves for different poles/zeros
- B.** that shows us how a closed-loop system behaves for different gains
- C.** that shows us how a closed-loop system behaves when we close our eyes

Quiz 1

calculating closed-loop pole:
 $s + k - 1 = 0 \Rightarrow s = 1 - k$

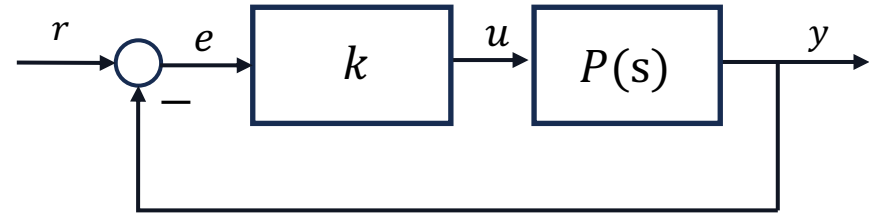
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$$\angle s = \frac{\angle(-k) \pm q \cdot 360^\circ}{n - m}$$

$$s_{com} = \frac{\sum_{i=1}^n p_i - \sum_{j=1}^m z_j}{n - m}$$



Quiz 2

Draw the Root Locus of the following transfer function: $G(s) = \frac{s+2}{(s+3)(s+4)(s+5)(s+6)}$

1. Calculate poles & zeros of the open-loop

$$z_1 = -2, p_1 = -3, p_2 = -4, p_3 = -5, p_4 = -6$$

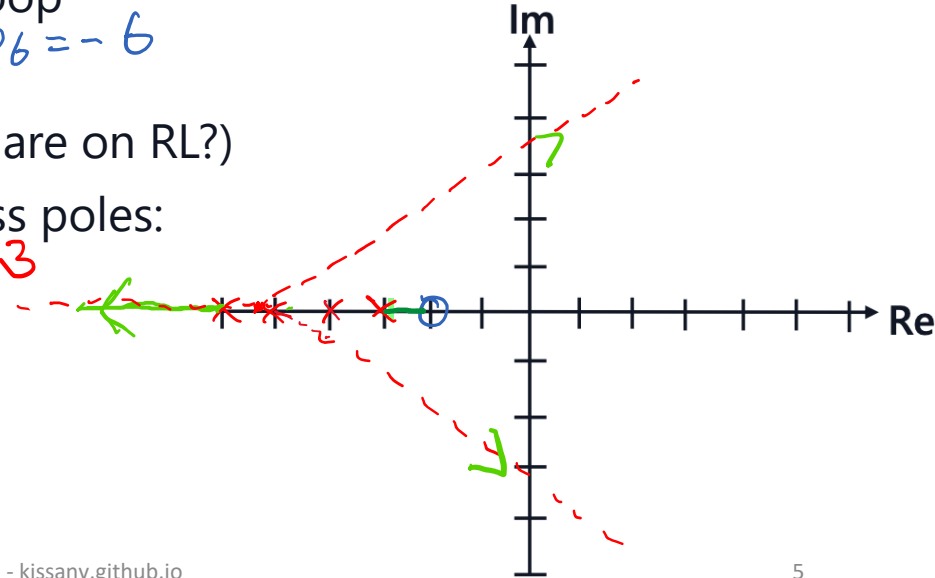
2. Analyze the real axis (which segments are on RL?)

3. Calculate the asymptotes for the excess poles:

$$s_{com} = \frac{1}{3}(-6-5-4-3+2) = -\frac{16}{3} \approx -5.3$$

$$\angle s = \frac{180^\circ \pm q \cdot 360^\circ}{3} = 60^\circ \pm q \cdot 120^\circ$$

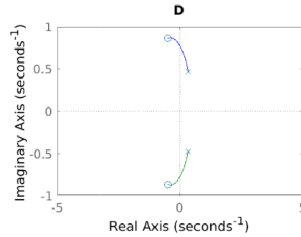
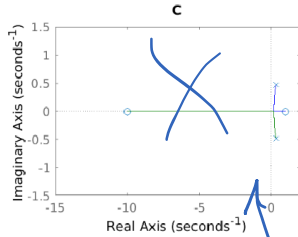
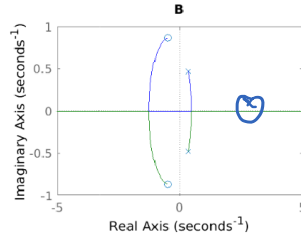
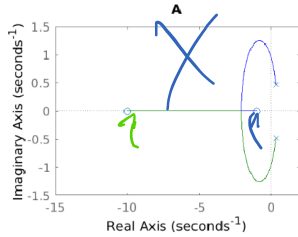
4. Draw Root Locus



Quiz 3

Problem: Given in Figure 4 are the root locus plots of four different transfer functions for $k > 0$.

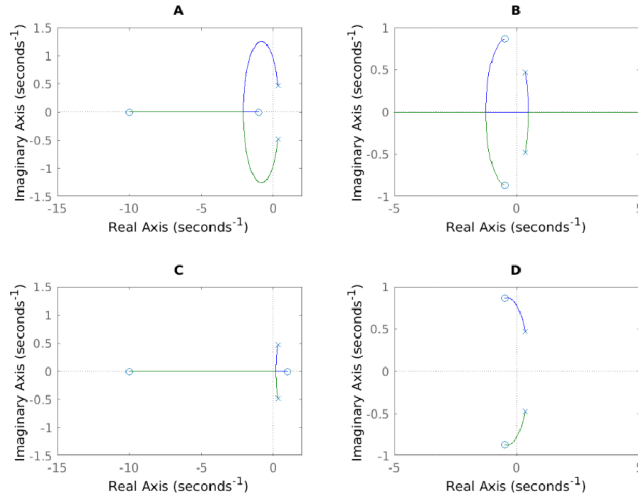
Q19 (1.5 Points) Assign each transfer function to the corresponding root locus plot.



Transfer Function	A	B	C	D
$L_1(s) = \frac{s^2 + s + 1}{3s^2 - 2s + 1}$				X
$L_2(s) = -\frac{s^2 + s + 1}{3s^2 - 2s + 1}$		X		
$L_3(s) = \frac{(s + 10)(s - 1)}{3s^2 - 2s + 1}$			X	
$L_4(s) = \frac{(s + 10)(s + 1)}{3s^2 - 2s + 1}$	X			

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$L_4(s) = \frac{(s + 10)(s + 1)}{3s^2 - 2s + 1}$	X			

Course Schedule

Subject		Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control	8
	Frequency response, Bode plots	9
	The Nyquist condition, Time delays	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

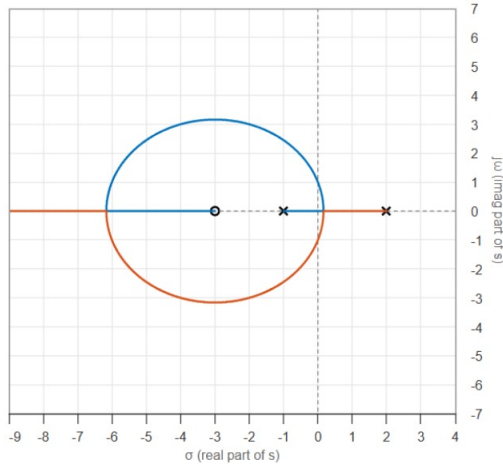
Today

1. System Specifications
2. Time-Domain Specifications
3. PID Control

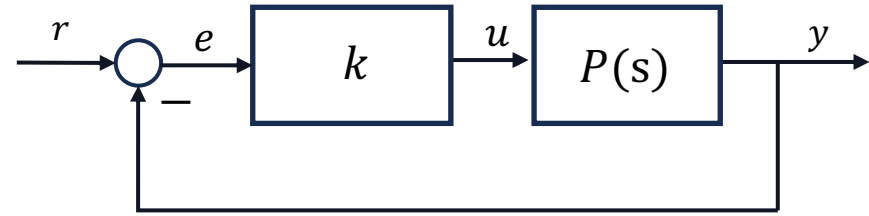
1. System Specifications

Introduction

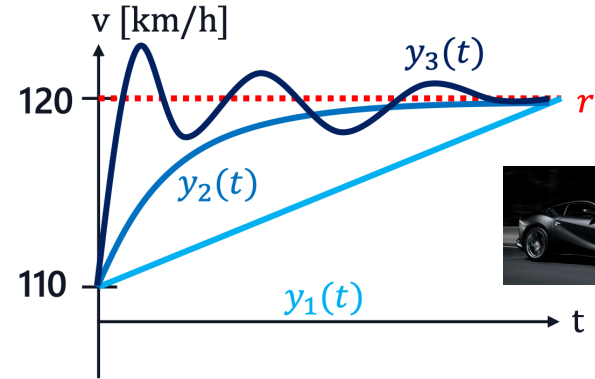
Week 7: Closed-Loop Stability



Job's finished? 🤔

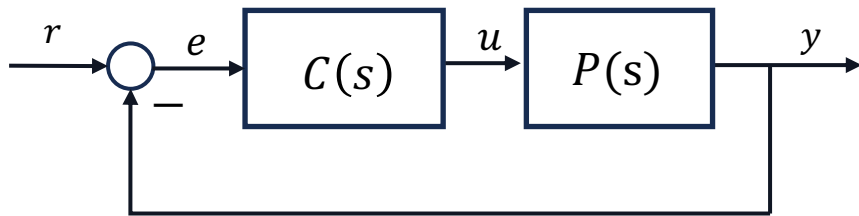


Week 8: Closed-Loop Specifications



We need to make sure it's quick and can handle disturbances and errors well!

Steady-State Error

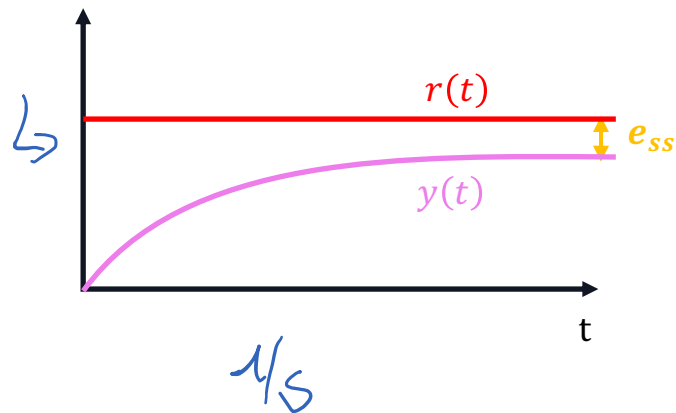


Assume we apply a step input to a system. Does the output approach the reference?

As defined last week, the relationship between r and e is captured with $S(s) = \frac{1}{1+L(s)}$

we find the steady-state error as:

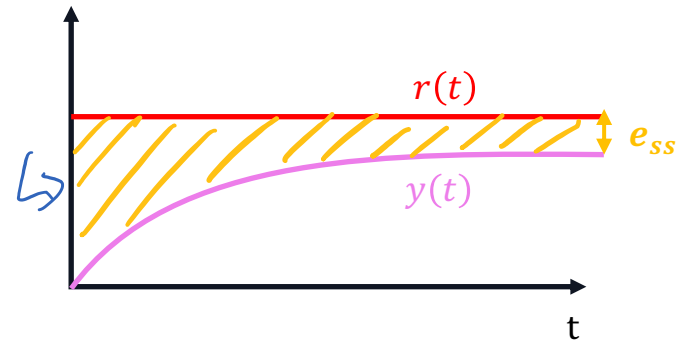
$$\begin{aligned}
 \hookrightarrow e_{ss} &= \lim_{t \rightarrow \infty} e(t) \stackrel{\text{using FVT}}{=} \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s S(s) R(s) \\
 &\text{step input } R(s) = \frac{1}{s}, \text{ hence} \\
 e_{ss} &= \lim_{s \rightarrow 0} s \cdot \frac{1}{1 + C(s)P(s)} \cdot \frac{1}{s} = \frac{1}{1 + C(0)P(0)}
 \end{aligned}$$



That means, that for e_{ss} to be zero $\rightarrow L(0) = C(0)P(0) = \infty$

$\rightarrow$ What property must the general $L(s)$ possess for this to hold true? We need a pole at $s = 0$!

Integrators



- If constant error e is integrated, it steadily grows
- Controller tries to minimize this growth and drives $e(t)$ to zero

Example: Given an open loop system: $L(s) = \frac{1}{s+2}$ what is e_{ss} for an step input?

$$e_{ss} = S(0) = \frac{1}{1 + L(0)} = \frac{1}{1 + 1/2} = \frac{2}{3}$$

$$e_{ss} = \frac{1}{1 + L(0)}$$

Let's add an integrator: $L(s) = \frac{1}{s} \cdot \frac{1}{s+2} = \frac{1}{s^2+2s}$ } $L(0) = 1/0$

$$e_{ss} = S(0) = \frac{1}{1 + L(0)} = \frac{1}{1 + 1/0} = 0$$

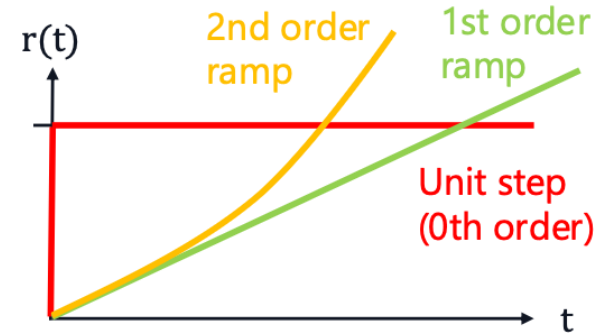
Great, now we know to make the output actually go where we want it to go!

q : order of ramp input
 Type: number of integrators

Steady State Error

e_{ss}	<i>order</i> $q = 0$	$q = 1$	$q = 2$
Type 0	$\frac{1}{1 + k_{Bode}}$	∞	∞
<i>q+1</i> Type <i>1</i>	0	$\frac{1}{k_{Bode}}$	∞
Type 2	0	0	$\frac{1}{k_{Bode}}$

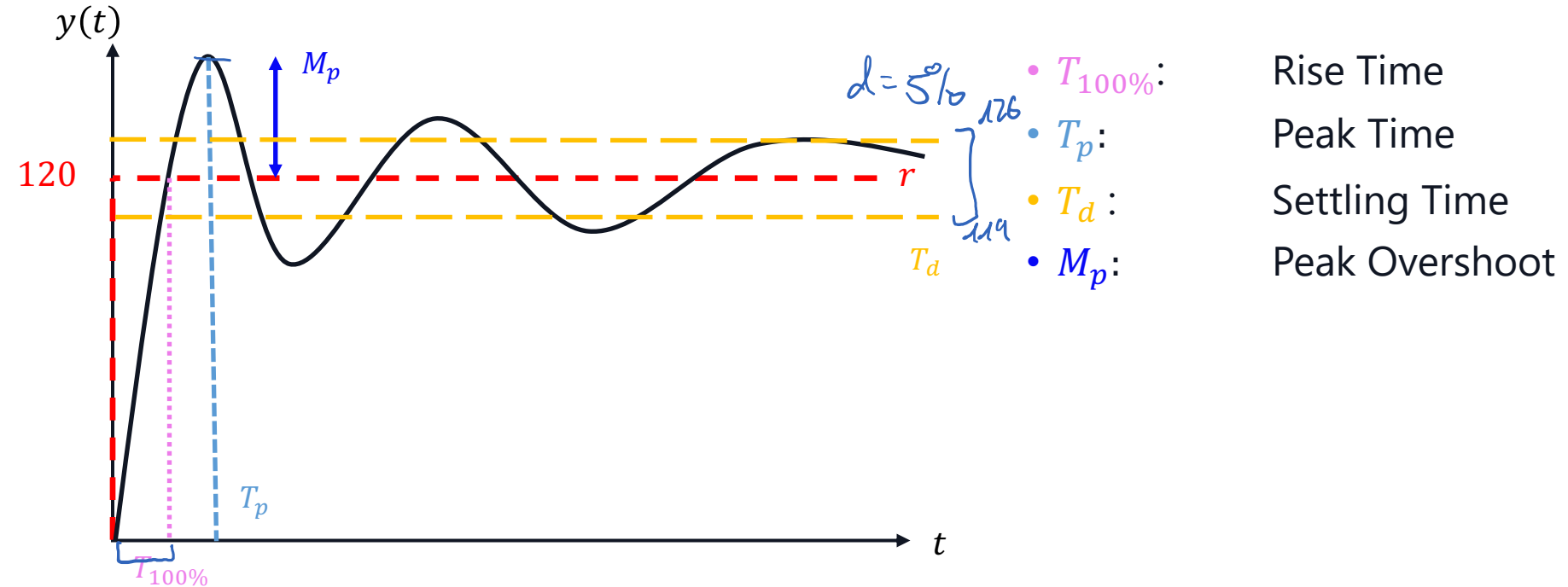
Control Systems 1 Lecture 8



- The steady-state error gets smaller as the (Bode) gain increases;
- For a zero steady-state error to ramps of order q , it is necessary to have at least $q + 1$ integrators on the path from the error e to the (reference/disturbance) input.

2. Time-Domain Specifications

Time-Domain Specifications



Step Response of a 1st Order System

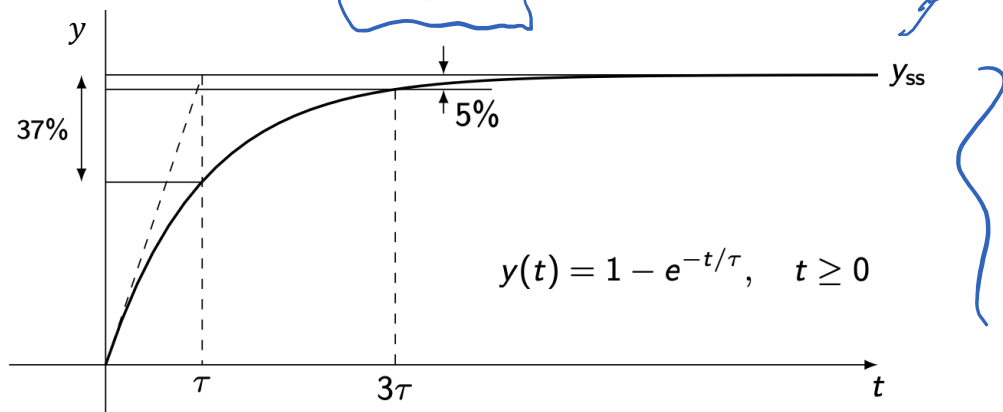
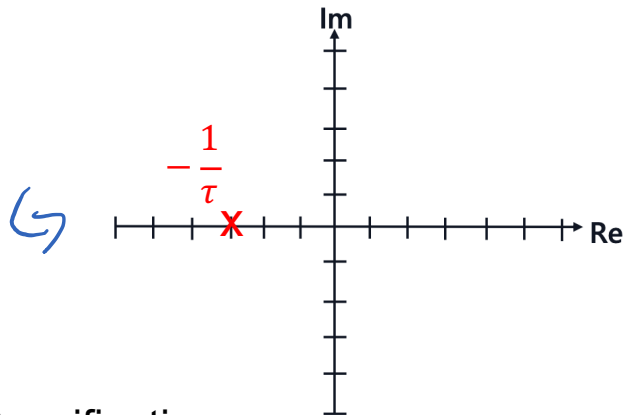
step response

$$G(s) = \frac{1}{\tau s + 1} \Leftrightarrow \left\{ \begin{array}{l} \dot{x} = -\frac{1}{\tau}x + \frac{1}{\tau}u \\ y = x; \end{array} \right.$$

$$x_0 = 0$$

$$Y(s) = G(s)U(s) = \frac{1}{\tau s + 1} \cdot \frac{1}{s}$$

$$\text{P.F.D: } Y(s) = -\frac{1}{s + \frac{1}{\tau}} + \frac{1}{s} \xRightarrow{\mathcal{L}^{-1}} y(t) = 1 - e^{-\frac{t}{\tau}}$$

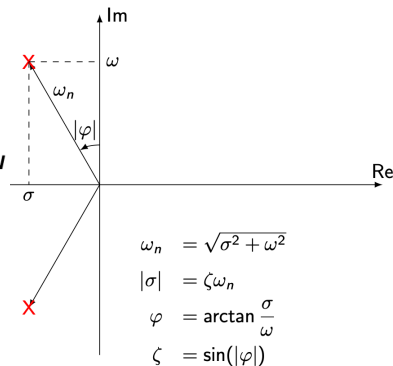


Specifications:

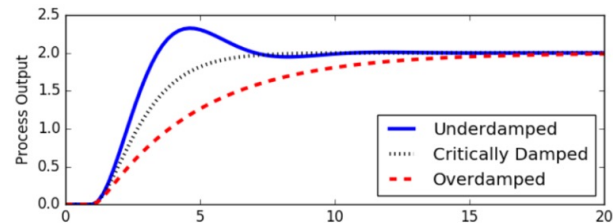
Settling Time:

Time to get within $d\%$ of tolerance of the reference signal

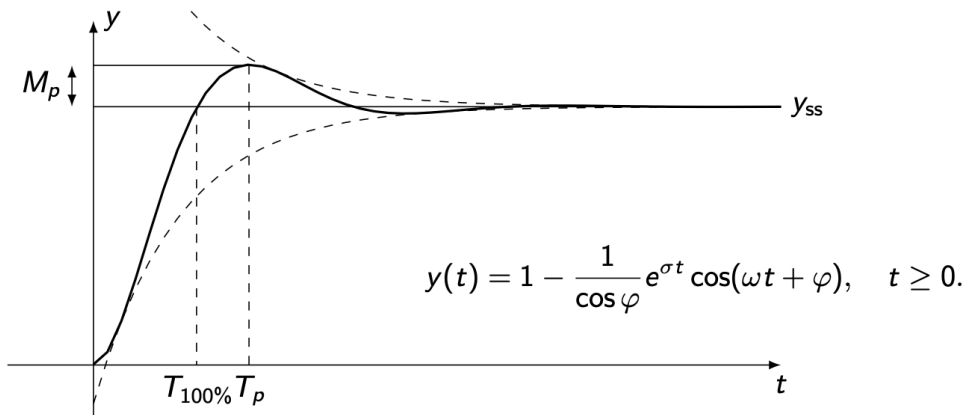
Step Response of a 2nd Order System

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \Leftrightarrow \begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} x + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} u \\ y = x; \end{cases}$$


$$\begin{aligned} \omega_n &= \sqrt{\sigma^2 + \omega^2} \\ |\sigma| &= \zeta\omega_n \\ \varphi &= \arctan \frac{\sigma}{\omega} \\ \zeta &= \sin(|\varphi|) \end{aligned}$$



For an underdamped system ($\zeta < 1$):



Specifications:

Settling Time: $T_d = \frac{1}{|\sigma|} \log\left(\frac{100}{d}\right)$

Peak Time: $T_p = \frac{\pi}{\omega}$

Peak Overshoot: $M_p = e^{\frac{\sigma\pi}{\omega}}$

Dampening Ratio: $\zeta^2 = \frac{\ln(M_p)^2}{\pi^2 + \ln(M_p)^2}$

Rise Time: $T_{100\%} = \frac{\frac{\pi}{2} - \varphi}{\omega} \approx \frac{\pi}{2\omega_n}$

Time-Domain Specifications Visualization

Specifications for 2nd order systems:

Settling Time:

$$T_d = \frac{1}{|\sigma|} \log\left(\frac{100}{d}\right)$$

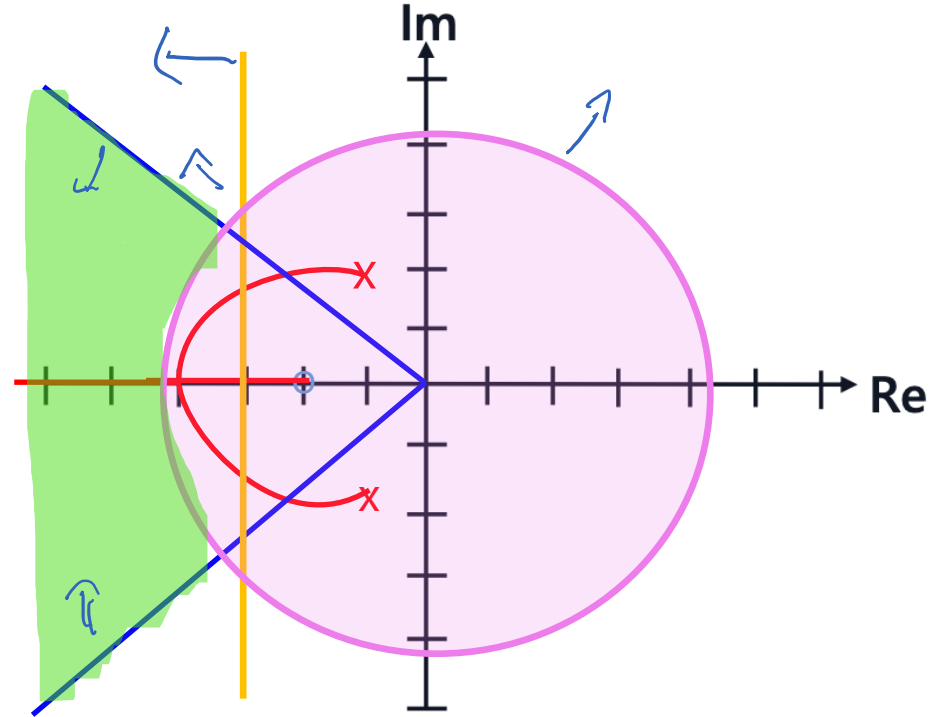
Peak Overshoot:

$$M_p = e^{\left(\frac{\alpha\pi}{\omega}\right)}$$

Rise Time:

$$T_{100\%} = \frac{\frac{\pi}{2} - \varphi}{\omega} \approx \frac{\pi}{2\omega_n}$$

Desired region

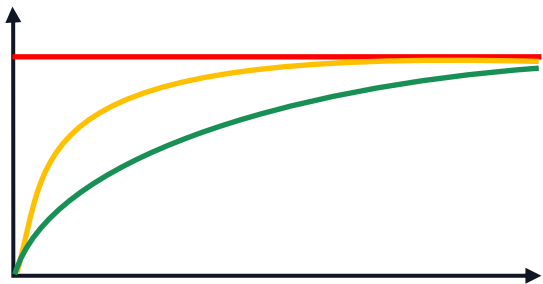


Dominant Poles Approximation

Thought experiment:

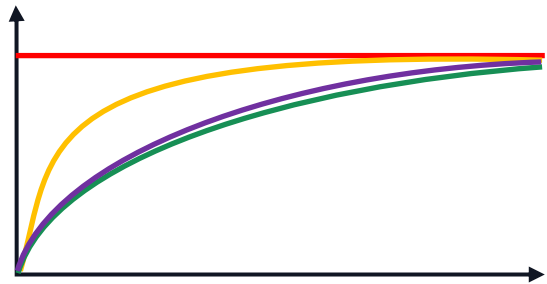
$$G_1(s) = \frac{1}{s+1}$$

$$G_2(s) = \frac{1}{s+5}$$



What do you expect for a system with both poles?

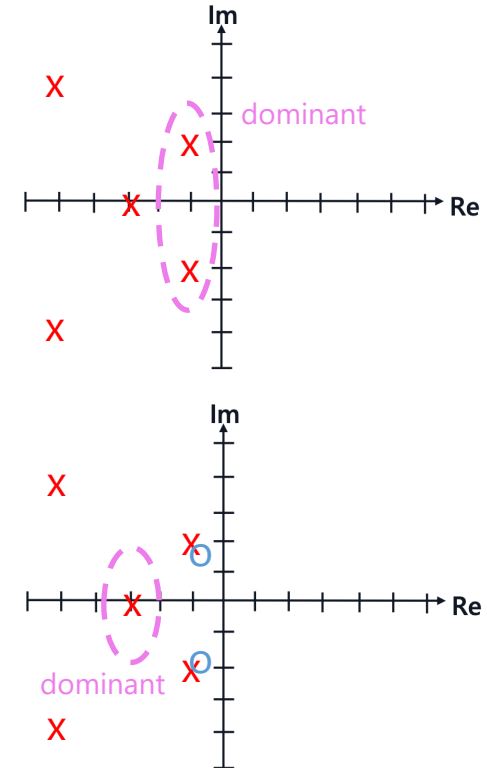
$$G_3(s) = \frac{1}{(s+1)(s+5)} = \frac{1}{s^2 + 6s + 5}$$



Poles closer to the origin have a higher influence!!

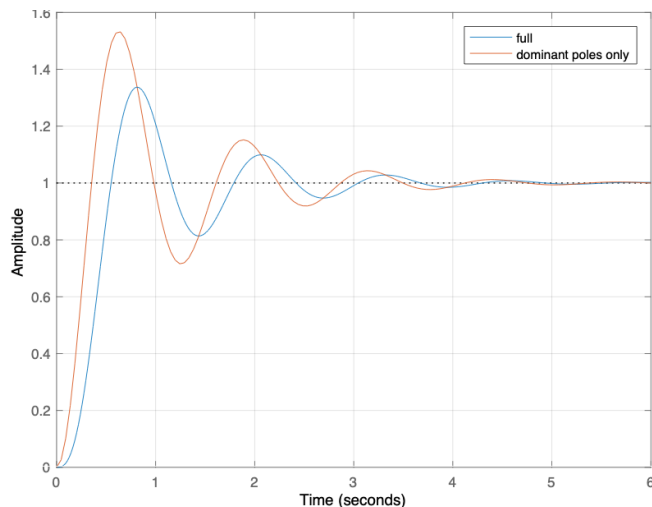
Dominant Poles Approximation

- Idea: Given a higher order system, approximate as a 1st order or 2nd order system, for which we have system specifications
- Recall Impulse Response:
$$y_{imp}(t) = \mathcal{L}^{-1}(G(s)) = r_1 e^{p_1 t} + \dots + r_n e^{p_n t}$$
- Poles close to imaginary axis decay the slowest $\rightarrow$ most dominant
- **Exception:** If those poles have small residues r_i due to zeros lying close to the poles (pole-zero cancellation)



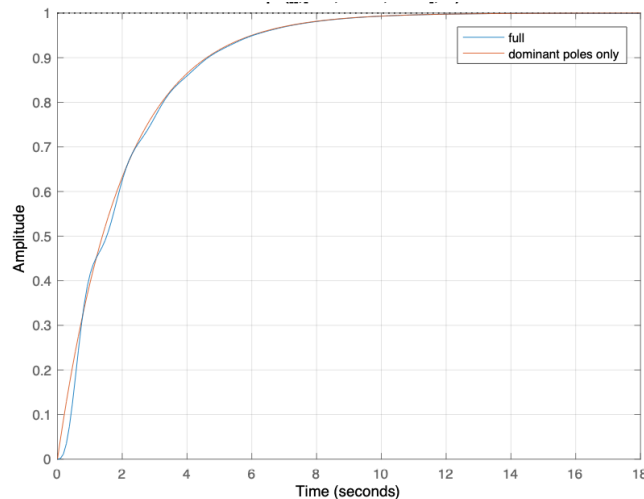
Dominant Poles Approximation

$$G_{approx}(0) = G(0)$$



$$G(s) = \frac{130}{(s+5)(s+1+5j)(s+1-5j)}$$

$$G_{dom}(s) = \frac{26}{(s+1+5j)(s+1-5j)}$$



$$G(s) = \frac{13}{(s+0.5)(s+1+5j)(s+1-5j)}$$

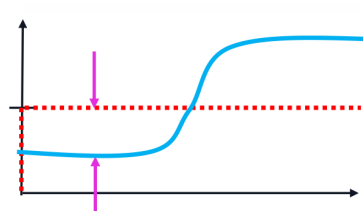
$$G_{dom}(s) = \frac{0.5}{s+0.5}$$

3. PID-Control

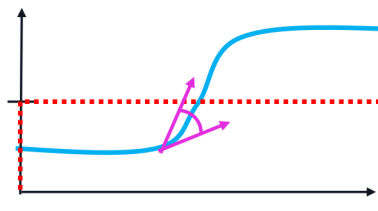
PID Control

$$u(t) = k_P e(t) + k_I \int_0^t e(\tau) d\tau + k_D \dot{e}(t),$$

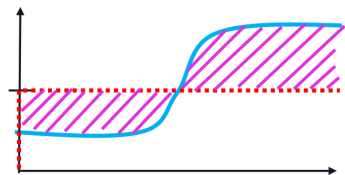
$$C(s) = k_P + \frac{k_I}{s} + k_D s = \frac{k_D s^2 + k_P s + k_I}{s}.$$



Proportional Gain:
Minimize distance to $r(t)$

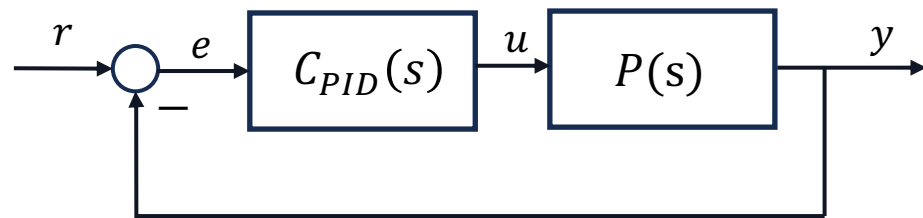


Derivative Gain:
Minimize "angle" to $r(t)$

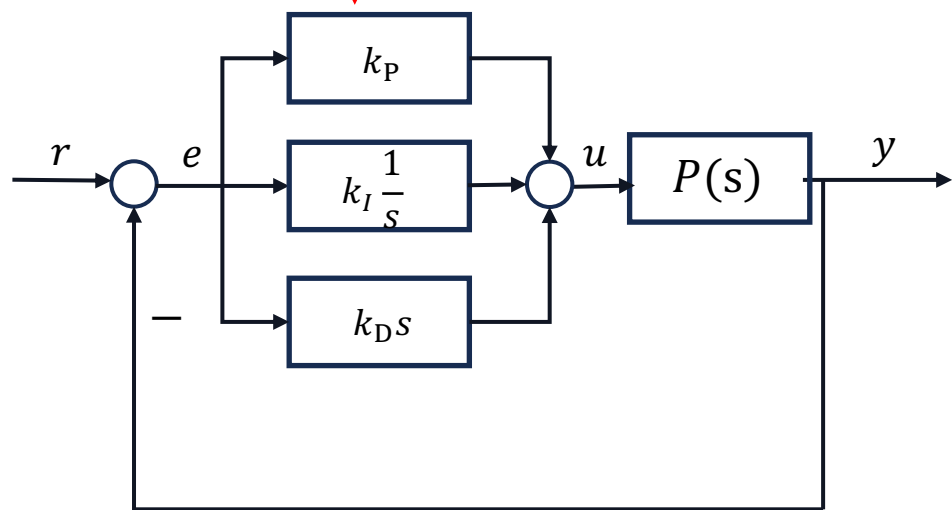


Integral Gain:

Minimize area to $r(t)$
graphs from Joel Tan



which really is just:



PID Effects

Proportional Control: $C(s) = k_p$

- Decrease the steady-state error
- Increase the closed-loop bandwidth
- Increase sensitivity to noise
- Can reduce stability margins for higher-order systems (2nd order or more)

Integral Control: $C(s) = k_I \frac{1}{s}$

- Decrease the steady-state error
- Increase the closed-loop bandwidth
- Increase sensitivity to noise
- Can reduce stability margins for higher-order systems (2nd order or more)

Derivative Control: $C(s) = k_D s$

- Reduce overshooting, increase damping
- Improves stability margins
- Increase sensitivity to noise

Video recommendation: <https://www.youtube.com/watch?v=UR0hOmjaHp0>

I personally would recommend solving at least the problems marked red

Tips for Problem Set 7

Easy	Medium	Hard
	1, 2, 3	

1(a): $T_{90} = (0.14 + 0.4\zeta) \frac{2\pi}{\omega_n}$. Apply the visualization of time-domain specifications

1(b): Apply Root Locus form to TF and sketch Root Locus

1(c): check if $T(s)$ stable, messy calculations, look at solution

2(a)-(c): Remember $G_{approx}(0) = G(0)$, calculate k_D by comparing poles from approximated TF with poles at $-5 \pm j$.

2(d): check solutions for the plot

3(a): here again remember the visualization of the time-domain specifications

3(b): from the time-domain specifications, remember how a 2nd order system is written:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

if unsure check solutions

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?
I would appreciate your feedback. Please let me know.

<https://docs.google.com/forms/d/e/1FAIpQLSdHI0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header>